

Simple Kenwood based requirments template - route is: User story and purpose -> Use case and marketing requirements -> technical requirements -> specs (partially execution)

1. Introduction and purpose of item or product

Device is intended to determine and log mainly the power consumption of small kitchen appliances while in real day use. The data will be analyzed and postprocessed to give statements about the power consumption, min, max, average, distribution in histograms to improve the devices and to evaluate if the life

2. Use case / User story / Marketing requirement

Req. ID	Detail	user requirement / tec req / spec	Source	Priority	Verification method	Standard / test method	Status	Notes / Evidence (firmware v6 — Shelly + OTA)
U 1	Device logs the power consumption of an attached slave device with common EU-plugs	user requirement	User	shall / should / can	Visual inspection / Demonstration / Test /		✔ Fulfilled	U1: 230 V switching delegated to certified Shelly Plug S MTR Gen3 (EU F-type, CE-marked). ESP32 receives push from shelly_push.js via POST /api/shelly_push. ShellyClient(ingest) parses & validates.
U 2	It stores the data permanently	user requirement	User	shall			✔ Fulfilled	U2: Samples stored on removable SD card as /log.csv in FILE_APPEND mode; no overwrite on boot (Logger.h).
U 3	The stored data is postprocessed	user requirement	User	shall			▲ Partial	U3: CSV format suitable for pandas/Excel. No dedicated post-processing scripts shipped yet — web UI provides /liveplot and /histogram for in-browser analysis.
U 4	The end user does not need to program or to set up the logger and just plugs it in	user requirement	User	shall			▲ Partial	U4: Plug-and-play after one-time factory pairing of the Shelly to SSD "PZEM_Logger". Once paired, the user only plugs both devices in. Degraded from ✔ to ▲ vs. v4 due to the Shelly pairing step; mitigated by pre-pairing at manufacture.
U 5	The use period ranges from 5 min to 50 h	user requirement	User	shall			▲ Partial	U5: Continuous operation supported (millis() rollover ~49.7 days). Long-run (>50h) not yet field-verified. Added risk vs. v4: the Wi-Fi link ESP32→Shelly must remain stable throughout.
U 6	It logs with appropriate sampling frequency and accuracy for standard small electric appliances such as hand mixers and jug blenders	user requirement	User	shall			▲ Partial	U6: Default 1 s push cadence from Shelly (was 0.5 s in v4 — Shelly meter updates at ~1 Hz). Adequate for kitchen appliances. Active power accuracy per Shelly datasheet. PF is derived (apower / V I), not hardware-measured — column renamed "pf_apparent" in CSV.
U 7	It can be safely used in the kitchen environment	user requirement	User	shall			⚡ Hardware	U7: Depends on enclosure build (IP44, cable glands, fuse, clearance/creepage). Not a firmware item - see specs section.
U 8	It does not cost (materials only) more than 60 €	user requirement	User	should			⚡ Hardware	U8: BOM-dependent. ESP32 + PZEM-004T + SD shield + housing are well below 60 € in single-unit pricing.
1	Device polls power per default every 1 s	tec req	U 6	shall			✔ Fulfilled	Req 1.0: INTERVAL_SHELLY_POLL_MS = 1000 ms (Config.h v6). Shelly pushes at this cadence; Logger::pollIfDue() consumes the cached value. Changed from 0.5 s (v4) — Shelly internal meter is 1 Hz.
2	Device logs power with affiliated timestep per default every 10 s to memory	tec req	U 6	shall			✔ Fulfilled	Req 2.0: INTERVAL_SD_FLUSH_MS = 10000 ms in Config.h. Logger::flushIfDue() writes RAM buffer to SD every 10 s. Each sample carries its own millis() timestamp.
3	Device does not overwrite existing log data	tec req	U 2	shall			✔ Fulfilled	Req 3.0: Log file is opened with FILE_APPEND (Logger::flushToSD). Reset only via explicit POST /reset.
4	The memory is removable / interchangeable	tec req		shall			⚡ Hardware	Req 4.0: Standard SD card on SPI - physically removable (PIN_SD_CS=25 etc.).
5	Device uses a format easy to post process on PC - for instance CSV or similar	tec req	U 3	shall			✔ Fulfilled	Req 5.0: CSV header "time,ms,voltage,V,power,W,pf_apparent" (LOG_FILE_HEADER in Config.h v6). Column renamed from "pf" to "pf_apparent" to reflect its derived nature.
6	memory resists 100 h which equals 36 000 k write cycles at 1/10s writing frequency	tec req	U 5 / U 2	shall			✔ Fulfilled	Req 6.0: SD cards withstand >100 k write cycles per block; 36 000 appends to one file is well within spec. Flush every 10 s (not 1/10 s), so the actual write load is 10x lower than the requirement.
7	It is capable to measure up to 10 A at 230 V	tec req	U 6	shall			⚡ Hardware	Req 7.0: PZEM-004T module spec is 100 A / 80-260 V (with included CT). Limited by external 8-10 A fuse per BOM.
9	Device transmits logged data via wireless function to host	tec req	U 4	should			✔ Fulfilled	Req 9.0: WiFi.mode(WIFI_AP_STA) — ESP32 runs softAP for phone/laptop; Shelly joins same AP as STA client. POST /api/shelly_push receives measurements every 1 s. No external network involved.
10	Allows to connect both types of electric devices: protective earth devices and EU-plug devices	tec req	U 1	shall			⚡ Hardware	Req 10.0: Depends on socket/plug wiring (grounded EU socket per BOM).
11	Fulfills the requirements for clearance and creepage	tec req	U 7	shall			⚡ Hardware	Req 11.0: Enclosure / PCB layout responsibility - not firmware.
12	electric protection is >= IP44	tec req	U 7	should			⚡ Hardware	Req 12.0: Enclosure responsibility (IP44 housing + cable gland per BOM).
13	allows safe flush of memory if the power connection is determined	tec req	U 7	should			✗ Not fulfilled	Req 13.0: No mains-loss detection implemented. PIN_VSUPPLY (GPIO 35) is configured as INPUT and ADC resolution set to 12 bit in setup(), but the value is never read. ACTION: poll ADC, on drop below threshold call logger.flushToSD() before backup capacitors deplete.
14	User interface: The device comes with a user interface that	tec req		N/A			! N/A	Section header (UI requirement group).
15	- blinks 1/s if polling and logging work correctly and no wireless connection is established	tec req		shall			✔ Fulfilled	Req 15.0: INTERVAL_LED_OK_MS = 500 ms toggles the LED at 1 Hz (StatusLed::update with LED_OK).
16	- blinks 5/s in case polling or logging do not work correctly	tec req		shall			✔ Fulfilled	Req 16.0: INTERVAL_LED_ERR_MS = 100 ms → LED_ERROR at 5 Hz. Triggered by SD error OR Shelly watchdog timeout (SHELLY_ERROR_THRESHOLD = 3 * 1000 ms = 3 s silence). Logger::ok() now checks shellyOk() & sdOk(). NOTE: LED will also blink 5 Hz during OTA upload — expected, recovers automatically on reboot.
16b	- blinks 0.5/s if polling and logging work correctly and wireless connection is established	tec req		can			✗ Not fulfilled	Req: 0.5 Hz blink when wireless client is connected is NOT implemented. StatusLed only has two states (LED_OK / LED_ERROR). ACTION: add a third state LED_OK_WIFI and switch on WiFi.isConnected() > 0.
17	Device comes with an additional on-line user interface once connected to host via wireless that	tec req		should			✔ Fulfilled	Req 17.0: Web portal served from softAP at default IP 192.168.4.1 (WebPortal::begin).
18	- displays the current life data on the host	tec req		should			✔ Fulfilled	Req 18.0: Home page polls /api/ive every 1 s; shows power (W), voltage (V), pf_apparent. JSON also includes shelly_ok, sd_ok, and ota_active fields. Label updated "pf" → "pf_apparent".
19	- allows to download all stored data - in the "DOWNLOAD DATA" section	tec req		should			✔ Fulfilled	Req 19.0: GET /download streams /log.csv as Content-Disposition: attachment (/WebPortal::handleDownload).
20	- allows to delete / reset the data storage memory - in the "DELETE MEMORY" section	tec req		should			✔ Fulfilled	Req 20.0: POST /reset calls Logger::resetSDFile() (deletes log.csv and re-creates with header). Confirmation dialog in JS.
21	- allows to set the poll rate between one poll every 1, 2, 5, 10, 30 s - in the "SETUP" section	tec req		can			▲ Partial	Req 21.0: /settings page rate grid allows {1000, 2000, 5000, 10000, 30000} ms. handleUpSettings(Save()) whitelists enforces this. Sub-second options (200/500 ms) removed — Shelly meter is 1 Hz. Status ▲, because rates other than 1/s are not yet validated end-to-end against the fixed 1 Hz Shelly push cadence.
22	- allows to set up the start of log at power consumptions greater than 1, 2, 5, 10, 20, 50 W	tec req		can			✔ Fulfilled	Req 22.0: Threshold grid {0, 1, 2, 5, 10, 20, 50} W in /settings UI. handleUpSettings(Save()) whitelists value. Logger::setPowerThreshold() applies immediately. Threshold lost on reboot (RAM only).
23	- allows the user to read a built in user manual - "MANUAL"	tec req		can			▲ Partial	Req 23.0: /readme page (PAGE_README) updated to reflect Shelly architecture and includes OTA section. Still light on troubleshooting depth — basic operations covered, no advanced setup guide.
24	- allows to plot the live data on the host - "LIVE PLOTTER"	tec req		can			✔ Fulfilled	Req 24.0: /liveplot canvas oscilloscope shows last 60 samples. fetchIntervalMs synced to /api/settings, floored at 1000 ms (Shelly meter cadence). Auto-scales Y axis to peak.
25	Device comes with a manual user interface in case the host is not connected via wireless	tec req		should			✗ Not fulfilled	Req 25.0: PIN_BUTTON (GPIO 33) is set as INPUT_PULLUP in setup() but never read anywhere. No manual UI exists.
26	- allows to reset and delete the data storage memory	tec req		should			✗ Not fulfilled	Req 26.0: No button-based reset (related to req 25.0).
27	device should carry all relevant information for wireless connectivt (ID, password...) on the device	tec req		should			✔ Fulfilled	Req 27.0: Housing labelled with WiFi name (PZEM_Logger), password (logger1234), web UI URL (http://192.168.4.1 or http://braun.pzem.local), and Shelly URL (http://192.168.4.2).
28	Device must not connect to the internet but locally only	tec req		should			✔ Fulfilled	Req 28.0: WiFi.mode(WIFI_AP_STA) — softAP only as anchor for Shelly and user. No WAN route. Shelly cloud disabled at factory. STA mode reserved for future NTP if needed.
28b	It is accessed by a domain that is plain text - BRAUN_PZEM - and not by hard to remember urls like: http://192.168.4.1	tec req		can			✔ Fulfilled	Req 28a.0: MDNS.begin(WIFI_AP_HOSTNAME) in WebPortal::begin(). Reachable at http://braun.pzem.local on iOS, macOS, and most Android. Previously open in v4 — code already existed.
29	Firmware updates can be installed over the air (OTA) without a physical USB connection to the ESP32	tec req	U 4	can	Demonstration / browser upload		✔ Fulfilled	Req 29.0: OTA implemented via ESP32 Arduino Update.h. GET /update serves an upload form (PAGE_OTA) with live progress bar; POST /update streams the .bin to Update.write() chunk-by-chunk and reboots on success. Logger::setOtaProgress(true) flushes the RAM buffer to SD and pauses polling before flashing — no log samples lost. Logging resumes automatically after reboot. SD-card retention of log.csv is unaffected (different memory).

4. Technical specs

	Use existing off the shelf components if possible - ESP32 - 230 V / 5 V Print AC/DC - SD card reader shield - grounded EU-socket - grounded EU-plug - empty off the shelf plastic housing - cable gland - LED+resistor - switch+resistor - wago spring terminals/connectors - overcurrent fuse 8-10A - AC power disconnection detection feature (CAPACITORS)	spec		can			▲ Partial	BOM: Shelly Plug S MTR Gen3 + certified 230~5 V PSU + ESP32 + SD shield + IP44 housing. Removed vs. v4: PZEM-004T, CT, EU socket, AC/DC print PSU, LED and pushbutton still pending in hardware build. Capacitor-based power-loss circuit still pending (Req 13).
	PIN layout ESP32: PinFunction GPIO 16 - UART2 RX (→ PZEM TX) GPIO 17 - UART2 TX (→ PZEM RX) GPIO 25 - SD CS GPIO 14 - SD MOSI GPIO 27 - SD CLK GPIO 26 - SD MISO GPIO 32 - LED GPIO 33 - Taster GPIO 35 - ADC Versorgungsbüberwachung	spec		can			✔ Fulfilled	Pin layout per Config.h v6: SD CS/MOSI/CLK/MISO = 25/14/27/26. LED = 32, ADC supply monitor = 35. GPIO 33 reserved for button (future). GPIO 16/17 (PZEM UART2) freed and unassigned. No 230 V connections anywhere on the ESP32 side.

Implementation status summary (firmware v6 — Shelly + OTA)

✔ Fulfilled	22
▲ Partial	7
✗ Not fulfilled	4
⚡ Hardware / out of firmware scope	7
! N/A (section header)	1